

CONVOLUTIONAL NEURAL NETWORKS For Autonomous Cars



V.P. Sampath is a senior member of IEEE and a member of Institution of Engineers India. He is currently working as technical architect at AdeptChips, Bengaluru. He is a regular contributor to national newspapers, IEEE-MAS section, and has published international papers on VLSI and networks

Convolutional neural networks (CNNs or ConvNets) used in machine learning are similar to neural networks. These are made up of neurons that have learnable weights and biases. Each neuron receives some inputs, performs a dot product and optionally follows it with a non-linearity. The whole network expresses a single differentiable score function from the raw image pixels on one end to class scores at the other, with a loss function (such as support vector machine/softmax) on the last (fully-connected) layer. Softmax function is just a generalisation of logistic function. It is used as a cost function for probabilistic multi-class classification, and by itself it is not a classifier.

CNNs have revolutionised the pattern-recognition computation process. Prior to the widespread adoption of CNNs, most pattern-recognition tasks involved hand-crafted features extraction followed by classification. With CNNs, features are now learned automatically from training examples. The CNN approach is especially powerful when applied to image recognition because convolution operation captures the 2D nature of images. By using the convolution kernels to scan an entire image, relatively few parameters need to be learned compared to the total number of operations.

CNN architecture

Unlike a regular neural network, the layers in a CNN have neurons arranged in three dimensions: width, height and depth. Here, 'depth' refers to the third dimension of an activation volume, not to the depth of a full

neural network, which can refer to the total number of layers in a network.

For example, input images in CIFAR-10 are an input volume of activations, and the volume has dimensions $32 \times 32 \times 3$ (width, height and depth, respectively). Neurons in a layer will only be connected to a small region of the layer before it, instead of all of the neurons in a fully-connected manner. Moreover, the final output layer for CIFAR-10 would have dimensions $1 \times 1 \times 10$, because by the end of the ConvNet architecture the full image will reduce into a single vector of class scores, arranged along the depth dimension.

Every layer of a CNN transforms the 3D input volume to a 3D output volume of neuron activations. In the example shown in Fig. 1, the red input layer holds the image, so its width and height would be image dimensions, and the depth would be 3 (red, green, blue channels).

Typically, CNN architectures comprise three types of layers stacked together: convolutional layer, pooling layer and fully-connected (FC) layer.

A simple ConvNet for CIFAR-10 classification has the architecture Input $\rightarrow$ Conv $\rightarrow$ Relu $\rightarrow$ Pool $\rightarrow$ FC as detailed below:

1. Input [$32 \times 32 \times 3$] holds the raw pixel values of the image; in this case, an image of width 32 and height 32, with three colour channels R, G and B

2. Conv layer computes the output of neurons that are connected to local regions in the input, each computing a dot product between their weights and a small region they are connected to in the input volume. This may result in a volume such as [$32 \times 32 \times 12$] if twelve filters are used

3. Relu layer applies an element-wise activation function, such as the $\max(0, x)$ thresholding at zero. This leaves the size of the volume unchanged ($[32 \times 32 \times 12]$).

4. Pool layer performs a down-sampling operation along the spatial dimensions

Fig. 1: Illustration of a ConvNet

